MICROPROCESSORS GROUP PROJECT PRESENTATION

GROUP 6

Drying Line Retractor

An automated system that detects rainfall and immediately retracts outdoor drying lines and protect clothes.

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SECTION 1

Problem Statement

The system addresses delayed manual response when rain begins unexpectedly

Core problem

- Delayed human response:

People may not be available or react quickly enough when rainfall starts, causing clothes to get wet before action is taken.

- User Inconvenience:

Constantly monitoring weather conditions or rushing to collect clothes during sudden rain is stressful and time-consuming.

- Unpredictable weather conditions:

Sudden rainfall, especially in tropical regions, makes manual intervention unreliable and ineffective.

- Risk of repeated washing and damage:

Wet clothes may develop odour or dirt again, leading to extra washing, water or energy waste, and fabric wear.

Failure chain without automation

Rain starts Sensor event appears → **Clothes exposed Fabric wets** → **Manual response Too late**

Design objective: detect rainfall early, decide instantly, and actuate both protective mechanisms before significant exposure occurs.

SECTION 2

Methodology

Functional approach used to sense rain and trigger mechanical protection

Approach to solving the problem

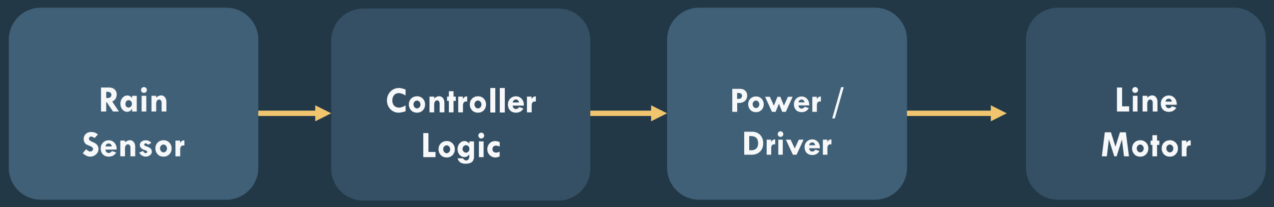
- A rain sensor continuously monitors the presence of water droplets.

- The controller compares the sensor state against a defined trigger threshold.

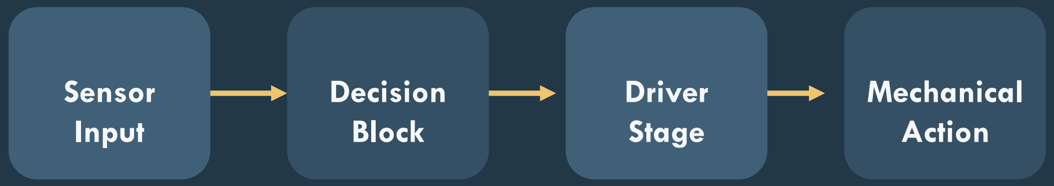
- When rain is detected, the controller sends commands to the servo motor.

- This servo retracts the drying line.

System architecture



Signal relationship



SECTION 4

Demonstration / Simulation - Software

This is a simulation using the Tinkercard software.

Simulation flow

1. Dry state
System idle

→

2. Rain trigger
Sensor changes

→

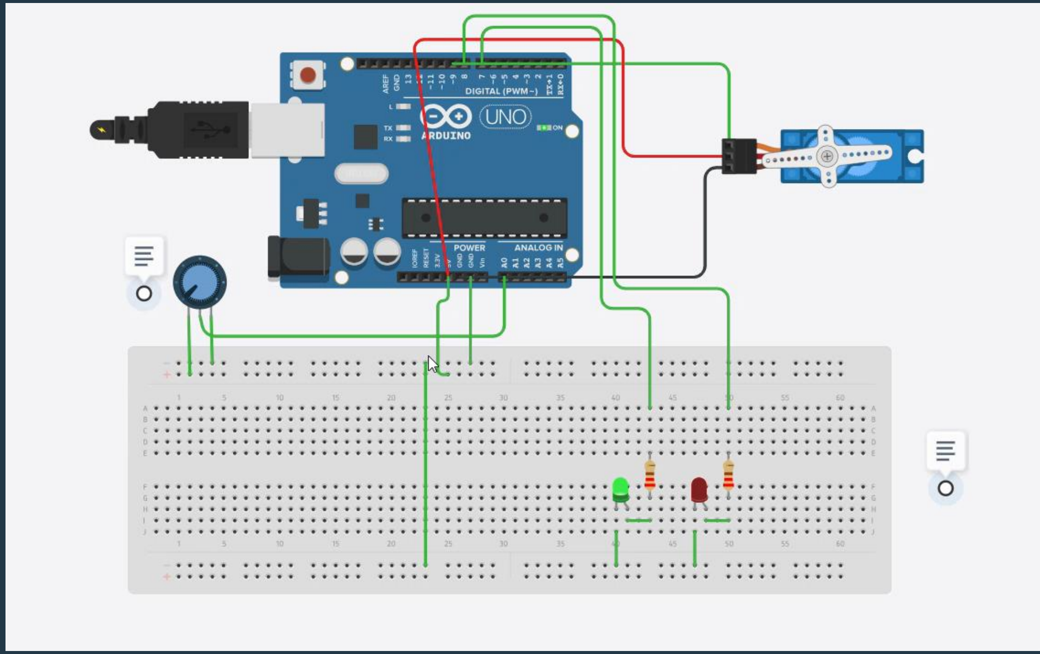
3. Protective action
By activating the servo

→

4. Safe end state
Line retracted.

- A potentiometer acts as a variable resistor, simulating a rain sensor by manually adjusting the voltage output to the Arduino's analog pin (A0). Turning the knob left lowers the reading below 400, simulating rain detected. Turning right raises it above 400, simulating dry conditions — identical in principle to how a real rain sensor responds to water.

- The red LED on means presence of rain, and the green LED being on means dry or absence of rain.



SECTION 4

Demonstration / Simulation - Hardware

Hardware implementation of the system employing the Arduino Uno microcontroller and moisture sensor

SECTION 5

Conclusion

Summary of findings, implementation challenges, and possible improvements

Findings

- The project provides a fast protective response to sudden rainfall.
- A single sensor-driven event can coordinate multiple household actions. Like closing of windows and other house apertures on the onset of the rain.
- The concept is practical for low-cost smart-home automation.

Challenges

- False triggers may occur without proper thresholding or debounce logic.
- Mechanical alignment and end-stop control are critical for reliable movement.
- Power delivery and actuator sizing must match the required load.

Possible improvements

- Add weather forecasting or IoT alerts for proactive protection.
- Integrate battery backup, enclosure sealing, and better status monitoring.
- Use real schematics, CAD mechanics, and measured test data in the final version.